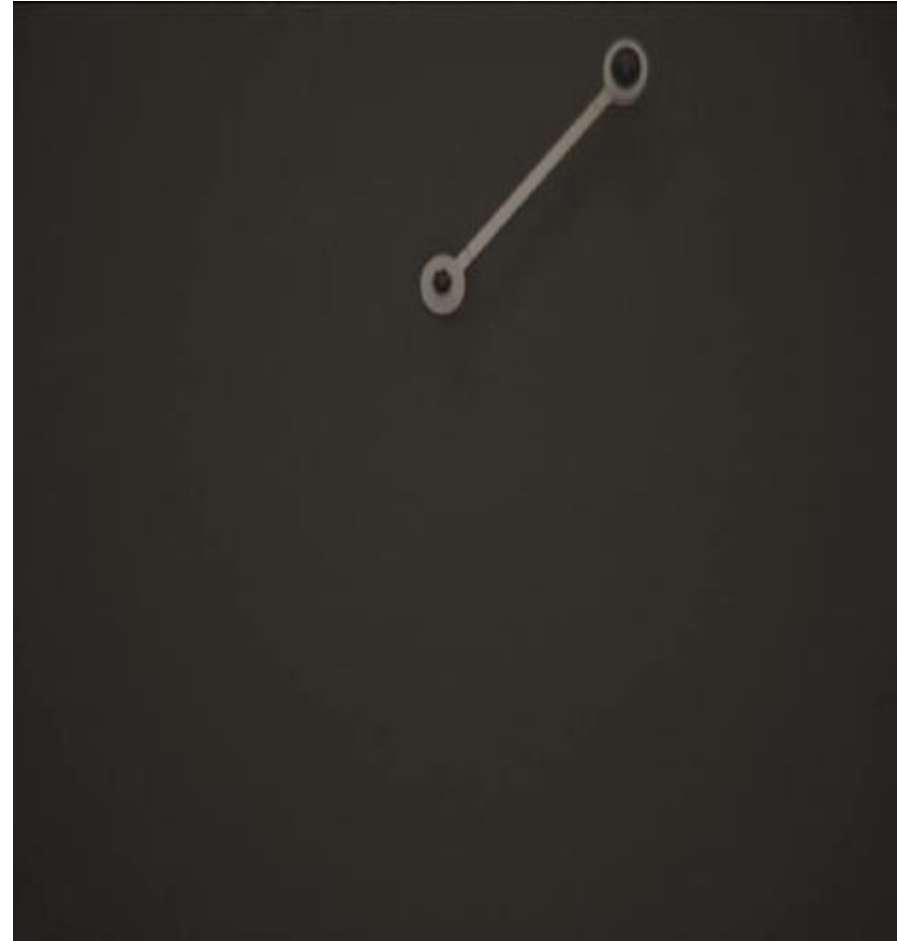


Lab 2: measuring a single pendulum

From high-speed video to a defensible model comparison

Physical model, finite-angle behavior, camera measurement, tracking, and Pre-Lab 2.



What you will do in Lab 2

Release pendulum from rest at three angle ranges.

Record motion with high-speed camera.

Track bob and convert $x(t)$, $y(t)$ into $\theta(t)$.

Compare period and frequency with model.



Physical system and coordinate

m : bob mass

L : pivot-to-bob-center distance

$\theta(t)$: angle from downward vertical

$\omega(t) = d\theta/dt$: angular velocity

Pivot remains fixed and every release begins with $\omega(0) = 0$.

Restoring torque and nonlinear equation

$$I \theta_{\text{double_dot}} = -m g L \sin(\theta)$$

$$\text{For a point mass: } \theta_{\text{double_dot}} + (g/L) \sin(\theta) = 0$$

Gravity drives the bob toward $\theta = 0$.

The sine term makes exact motion nonlinear.

Mass cancels in the ideal point-mass model.

Small-angle model and prediction

$\sin(\theta)$ approximately θ , for θ in radians

$$\omega_n = \sqrt{g/L} \quad f_n = \omega_n / (2\pi) \quad T_0 = 2\pi \sqrt{L/g}$$

Linear model predicts a sinusoidal angle history.

Longer L lowers natural frequency.

Use model mainly near equilibrium.

Why large initial angle changes period

At large angle, $\sin(\theta)$ differs from θ .

Exact restoring torque is weaker than linear model predicts at the same angle.

Finite-angle pendulum takes longer to complete a cycle.

Three Lab 2 releases reveal this change.



Energy during a swing

$$E = \frac{1}{2} m L^2 \omega^2 + m g L (1 - \cos(\theta))$$

At release θ is largest and $\omega = 0$.

Near bottom $\theta = 0$ and speed is largest.

Real data decays slowly due to air drag and pivot friction.

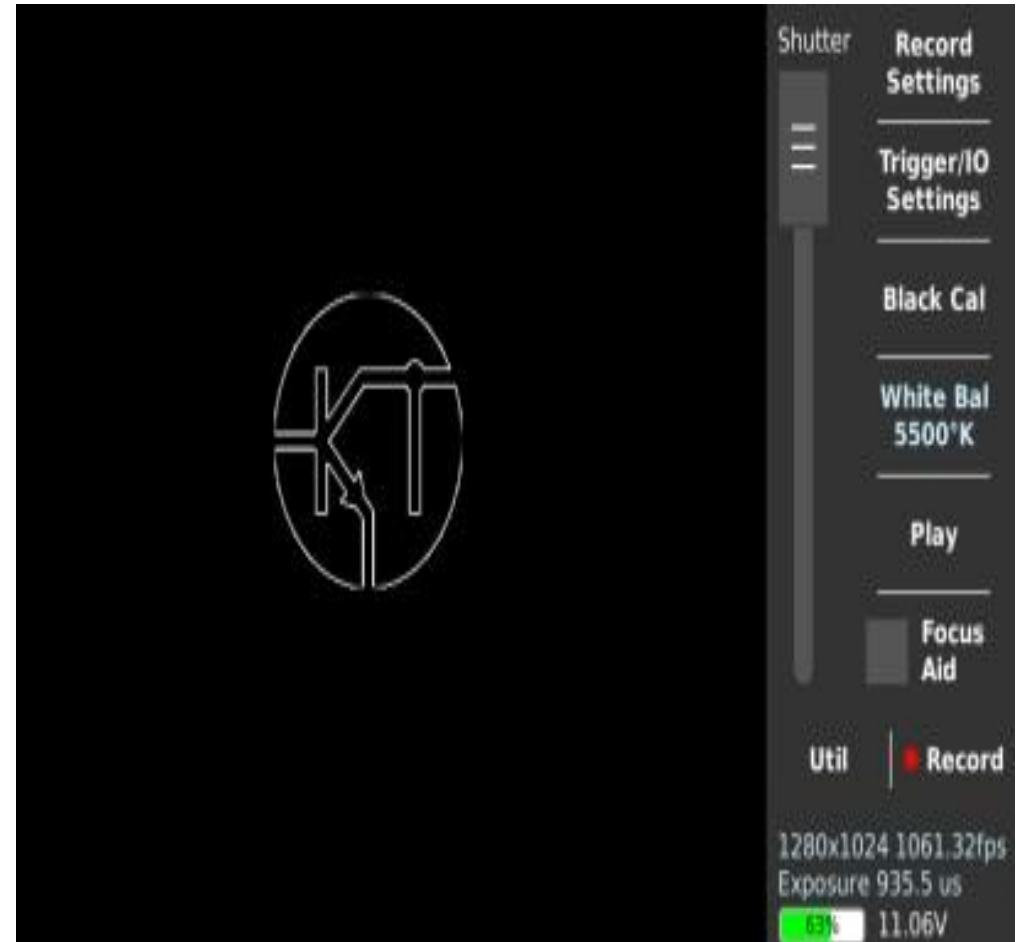
What the camera actually measures

Camera stores frames, not angle directly.

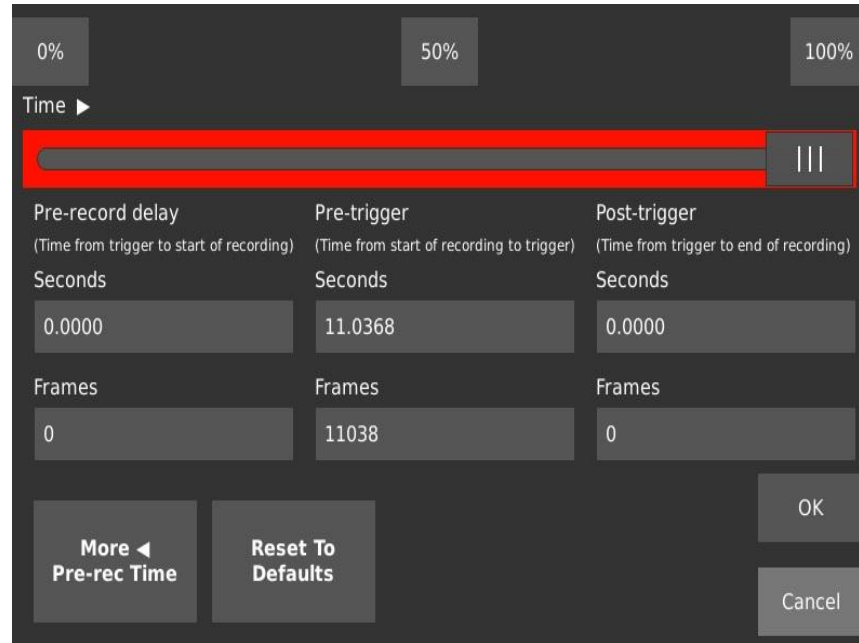
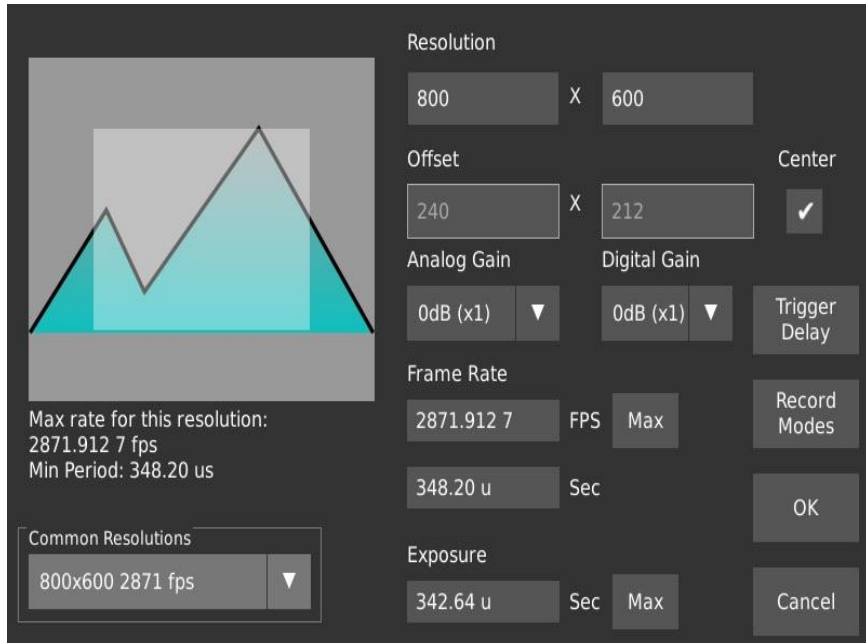
Every frame contains pivot and bob positions.

Useful record needs full motion in frame, sharp contrast, and visible bob center.

Frame rate becomes time spacing of measurement.



Camera settings and useful video



Record actual frame rate.

Use trigger delay that includes release and cycles.

Test framing before experiment.

Smooth video does not prove accurate tracking.

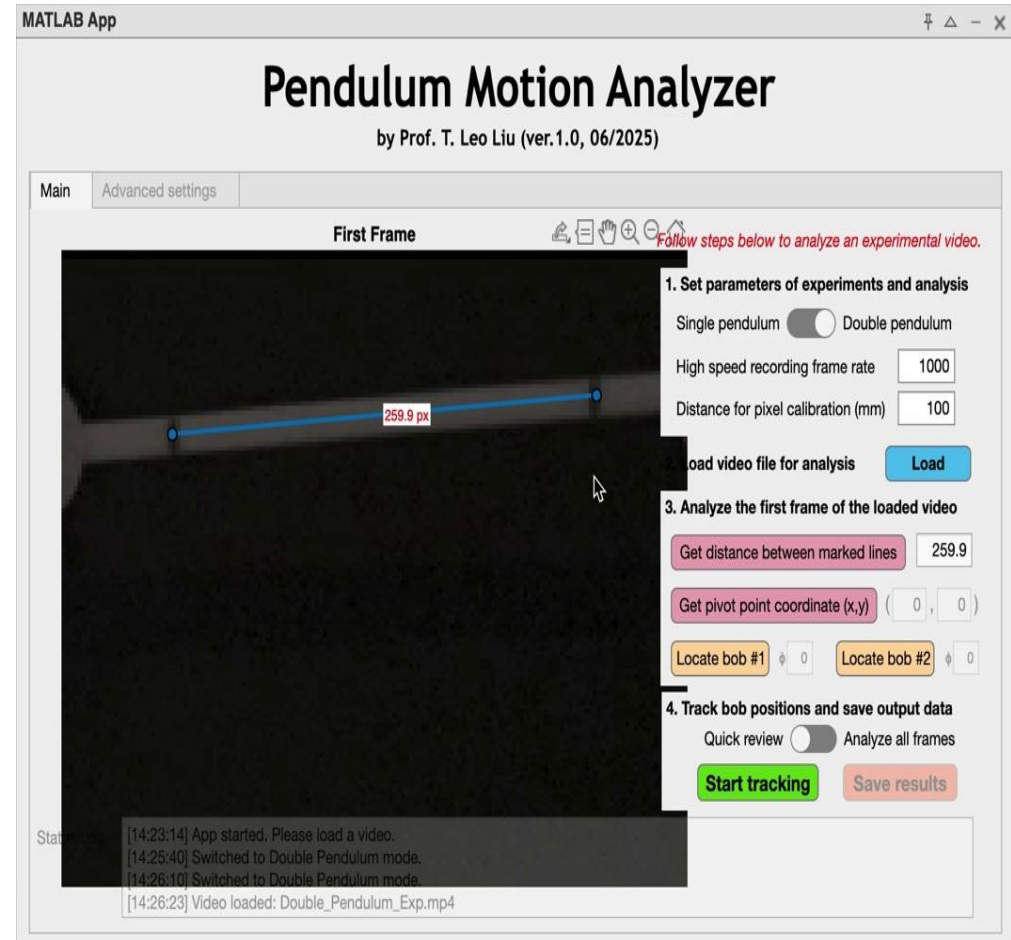
From pixels to coordinates

Calibration marks have known separation, normally 100 mm.

Calibration gives pixel-to-millimeter scale.

Do not move camera or change zoom after calibration.

App saves time and bob coordinates.



From coordinates to a signed angle

$$\text{theta}(t) = \text{atan2}(x_b - x_p, y_b - y_p)$$

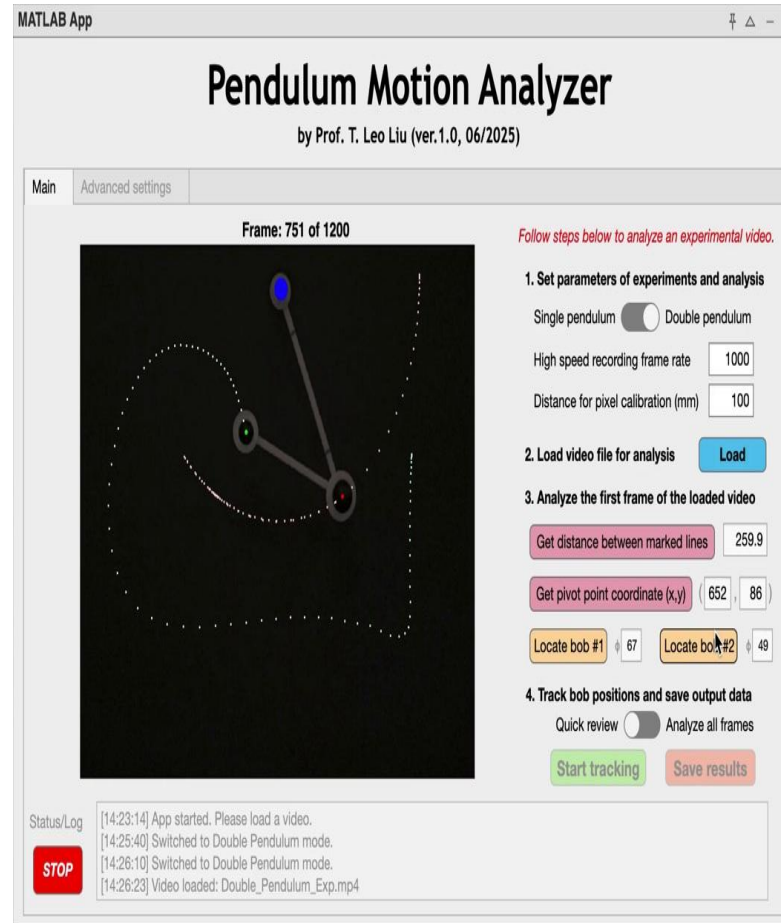
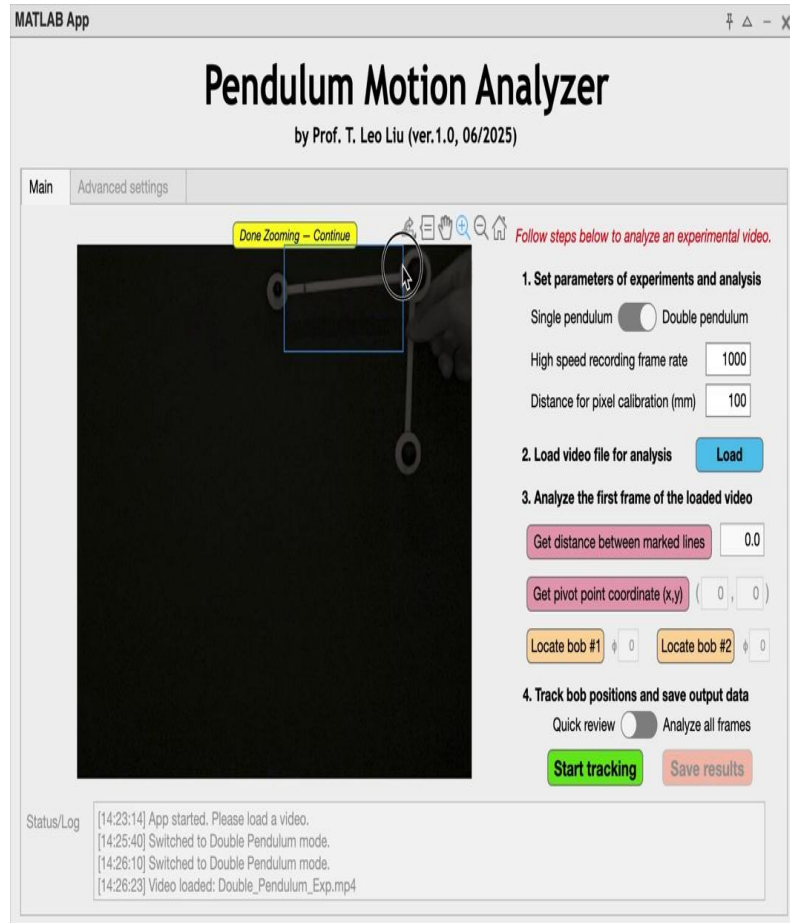
x_p, y_p are pivot coordinates; x_b, y_b are bob-center coordinates.

atan2 preserves the correct quadrant.

Use radians for modeling and derivatives.

A drifting pivot creates false angle change.

Tracking workflow in MATLAB



Select Single pendulum.

Enter actual camera frame rate.

Calibrate, select pivot and bob, then Quick Review.

Run Analyze All Frames only after marker follows bob.

Why frame rate belongs in report

At 1000 frames/s: Nyquist frequency = 500 Hz

For $L = 0.82$ m, one small-angle period has about 1800 frames.

Dense frames support time histories and derivatives.

Wrong frame rate gives wrong time, frequency, and angular velocity.

Angular velocity and noise

$$\omega(t) = d \theta(t) / dt$$

Differentiation magnifies frame-to-frame error.

Inspect $\theta(t)$ before interpreting $\omega(t)$.

State smoothing method and show it does not hide tracking problem.

Maximum speed occurs near bottom.

Period from time history and spectrum

Time domain: measure time between same-direction peaks.

Frequency domain: remove mean, compute FFT, identify dominant peak.

Bin spacing $\Delta f = 1/T_{\text{record}}$.

Time-domain period and spectral frequency should agree.

$$f = 1/T$$

$$\Delta f = 1/T_{\text{record}}$$

Lab 2 trial matrix

Trial A: θ_0 below 10 degrees. Closest test of small-angle theory.

Trial B: θ_0 from 10 to 90 degrees. Observe period change.

Trial C: θ_0 from 90 to 180 degrees. Expect strong nonlinearity; keep bob in frame.

All releases start from rest. Record θ_0 from protractor.

Data quality before accepting video

Bob remains in frame.

Bob edge and center are visible without glare.

Pivot and calibration marks are visible.

Release begins from rest.

Quick Review follows bob for nearly all frames.

Notes include file name, angle, frame rate, calibration distance.

Pre-Lab 2: what you submit

One Word or PDF before laboratory section.

Task 1: calculate ω_n , f_n , T_0 .

Task 2: predict behavior before simulation.

Task 3: estimate periods and explain finite-angle behavior.

Tasks 4-6: explain tracking, frame rate, and lab plan.

Evidence for combined Lab 2 and 3 report

Table: L , mass, θ_0 , predicted and measured frequency.

Plots: $\theta(t)$ and $\omega(t)$ with units.

Spectrum: meaningful range and stated bin spacing.

Discussion: distinguish finite-angle limitation from tracking error and damping.

In-class prediction

For $L = 0.82$ m, which release has longest period?

- A. 8 degrees
- B. 35 degrees
- C. 105 degrees

Explain using exact sine restoring torque.

Before leaving laboratory

Save and back up every video and MAT output.

Verify tracking for every trial.

Record initial angle, frame rate, calibration distance, length, mass, and file names.

Check one $\theta(t)$ plot and spectrum with GTA.

Keep raw data for combined report.

Key ideas for Lab 2

Camera images become positions; analysis converts positions to angle.

Small-angle theory has a clear range of validity.

Initial angle affects real pendulum period.

Frame rate and calibration are measurement settings.

Compare theory, time history, and spectrum.